SPEC CPU® 2026
What's New, What's Improved and Why it Matters

SERVER



David Reiner

SPEC President

Outline

What is SPEC?

CPU Importance in the AI Age

SPEC CPU Background

SPEC CPU 2026

Call to Action



Standard Performance Evaluation Corporation (SPEC®)

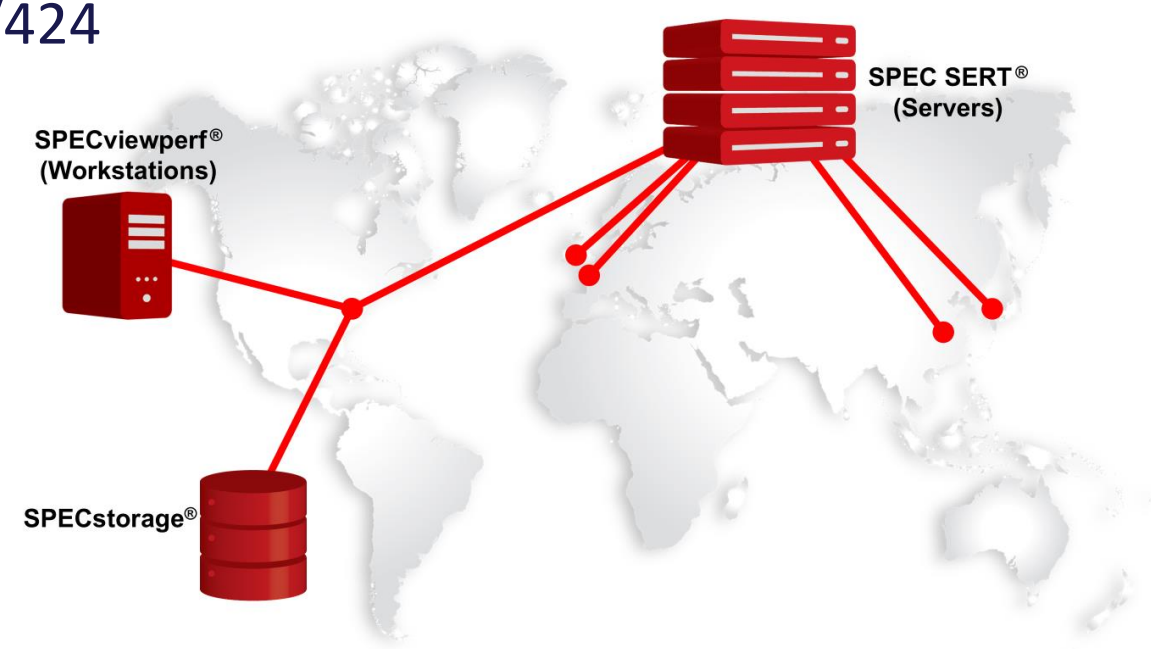
- An **industry led**, worldwide non-profit consortium **formed in 1988** to establish standards for measuring computer **performance and power**
- More than 115 member organizations from industry and academia, including 26 Asian organizations
- Committed to **vendor neutrality**, SPEC is the **trusted global leader** in computing benchmarks
- Through member collaboration with a focus on **real-world applicability**, SPEC offers more than 35 industry-standard benchmarks and tools for system performance evaluation in a variety of application areas
- Sponsors **research** and international **conferences** (ICPE) addressing diverse aspects of performance, including OCP China Day

SPEC Benchmarks

- SPEC currently publishes 35 benchmark suites used to measure systems from micro controllers to supercomputers
- Highlights include:
  - CoreMark®, IoTMark®, and AudioMark™ for Micro-processors
  - SPECviewperf®, SPECworkstation®, and SPECapc® for Graphics and Workstations
  - SPECstorage® for Disk and Network File Systems
  - SPECcloud® for Data Centers
  - SPECjbb® and SPECjvm® for Java Virtual Machines (JVM)
  - SPEC VIRT® for Virtualization Environments
  - SPECpower_ssj® to measure Power Efficiency
  - SPEC CPU® for General-purpose Computers
  - SPEChpc® for Supercomputers

Government Energy Efficiency Regulations

- SPEC partners with governments and standards bodies across the globe to provide energy efficiency benchmarks for use in regulations and programs
- The SERT suite is the standardized energy benchmarking tool for:
  - EU Lot 9 Ecodesign Regulation (EU) 2019/424
  - US ENERGY STAR® Version 4.0 for Servers
  - Japan's Top Runner Server Regulation
  - ISO/IEC 21836:2020
  - ETSI EN 303 470



Do Server CPUs and Benchmarks Matter Today?

- The proliferation of server CPU instruction set architectures (ISA)s and vendors increases the need for high-quality performance and power benchmarks
- Agentic AI has significantly increased the need for and importance of server CPUs



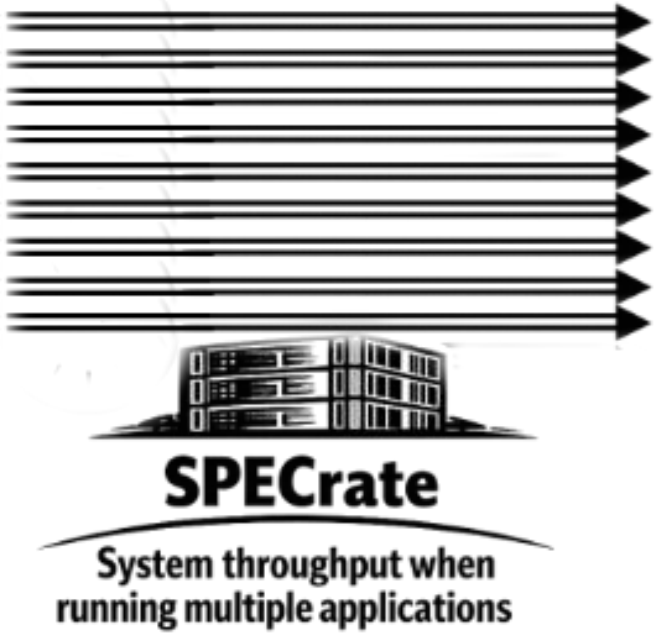
What Does SPEC CPU® Measure?

Measures the performance of

- General-purpose processors (CPU)
- Memory hierarchy
- Compilers
- Power Efficiency (optional)

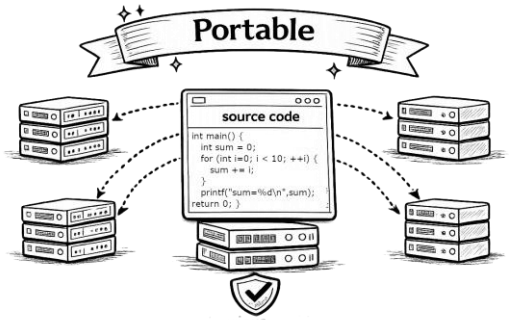
Composed of four suites

- SPECrate® Integer
- SPECrate® Floating Point
- SPECspeed® Integer
- SPECspeed® Floating Point





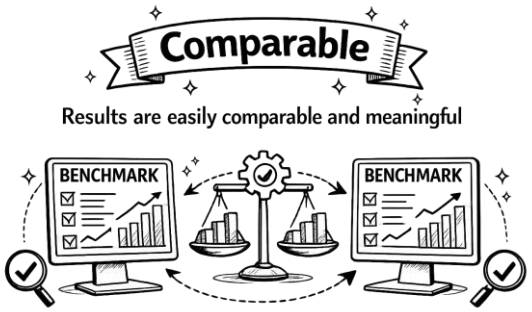
Seven Design Principles for SPEC CPU



Able to target many architectures and platforms without bias



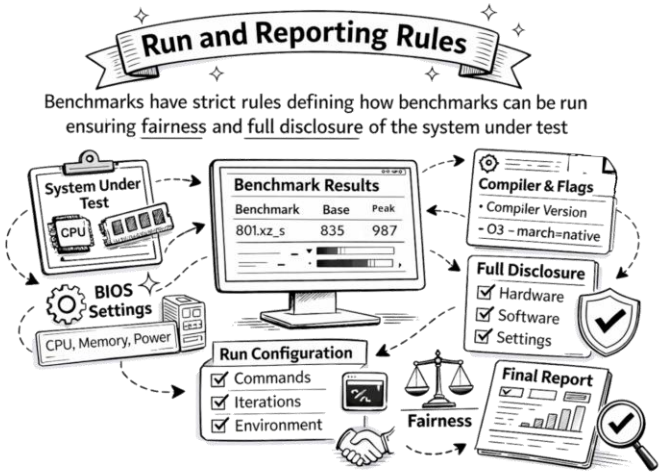
Others can repeat your test and verify your performance claim



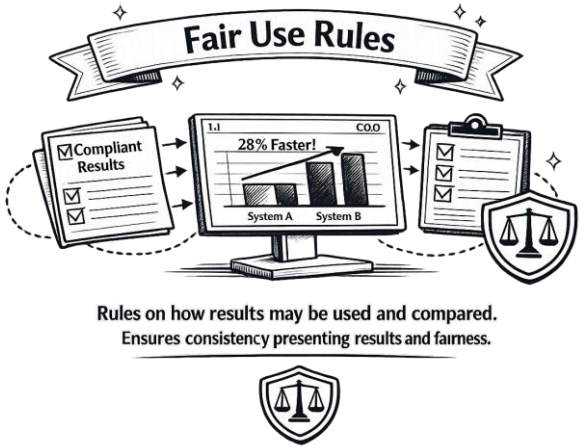
Results are easily comparable and meaningful



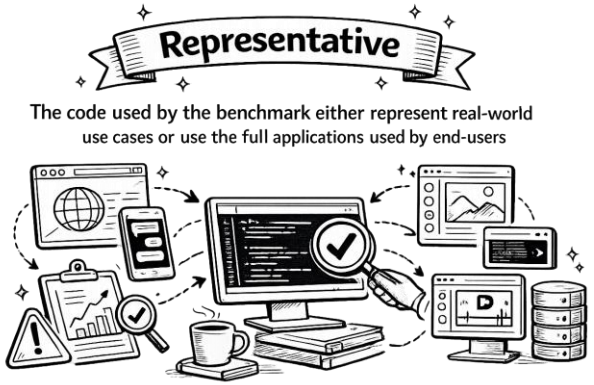
Performance doesn't matter if the results are wrong



Benchmarks have strict rules defining how benchmarks can be run ensuring fairness and full disclosure of the system under test

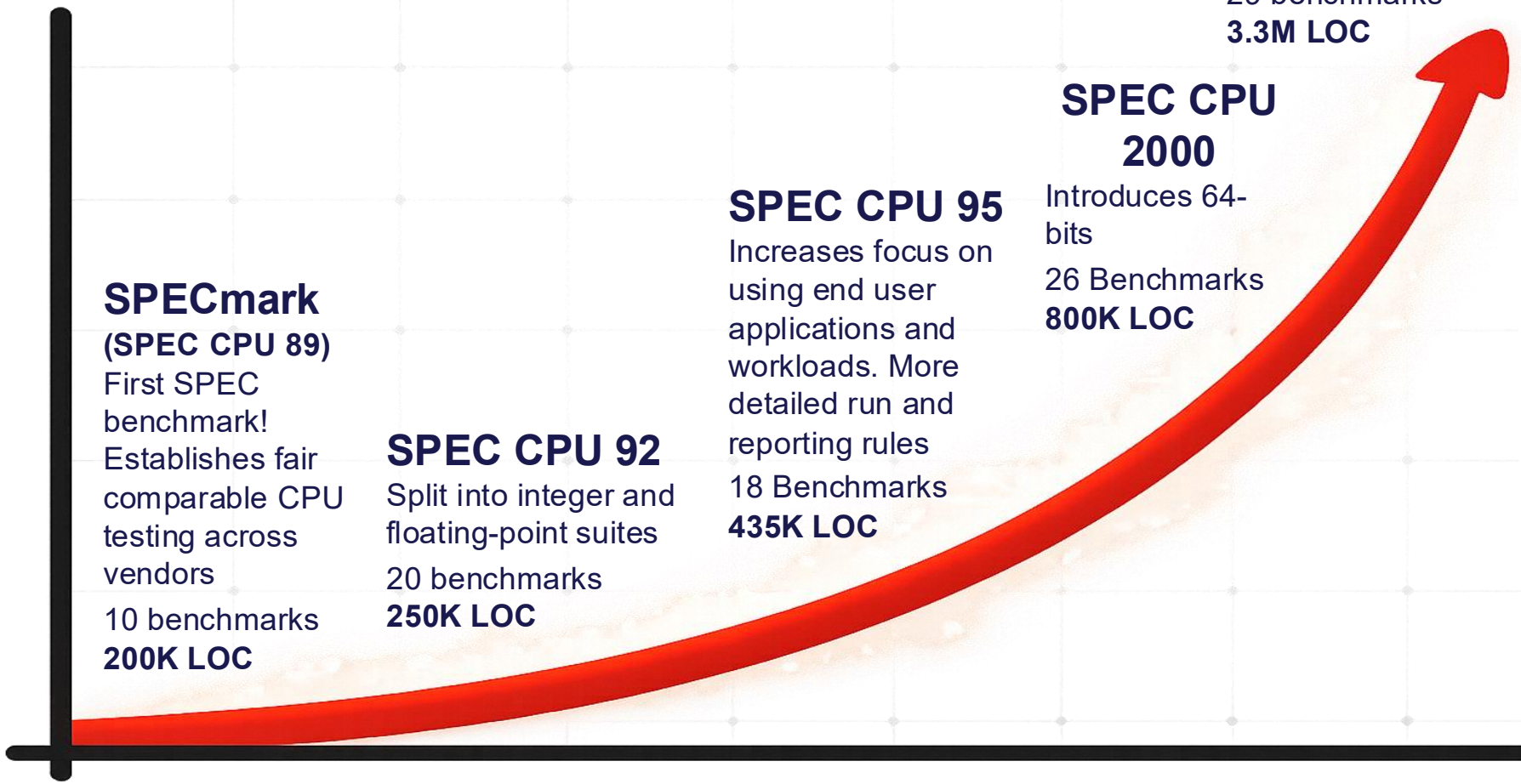


Rules on how results may be used and compared. Ensures consistency presenting results and fairness.



The code used by the benchmark either represent real-world use cases or use the full applications used by end-users

SPEC CPU History



SPECmark
(SPEC CPU 89)
First SPEC benchmark! Establishes fair comparable CPU testing across vendors
10 benchmarks
200K LOC

SPEC CPU 92
Split into integer and floating-point suites
20 benchmarks
250K LOC

SPEC CPU 95
Increases focus on using end user applications and workloads. More detailed run and reporting rules
18 Benchmarks
435K LOC

SPEC CPU 2000
Introduces 64-bits
26 Benchmarks
800K LOC

SPEC CPU 2006
Larger data sets and more diverse applications
29 benchmarks
3.3M LOC

SPEC CPU 2017
Adds explicit parallelism and power efficiency metrics
43 benchmarks
7.1M LOC

SPEC Published results

46,247+	SPEC CPU 2017
48,381	SPEC CPU 2006
7,654	SPEC CPU 2000
2,574	SPEC CPU 95
1,199	SPEC CPU 92
277	SPEC CPU 89

What's New in SPEC CPU 2026?

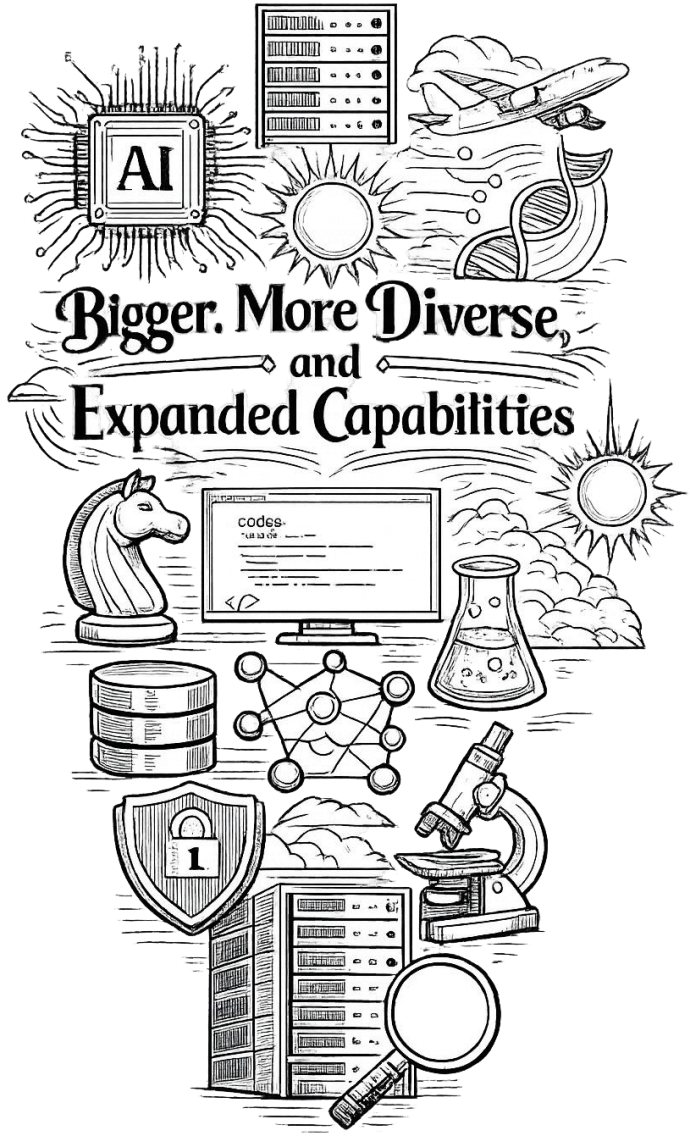
- Many new modern applications and domains
- 52 benchmarks, up from 43
- 16.7 million lines of source code, up from 7.1 million
- Broader coverage of microarchitecture behaviors
- Updated language standards: C18, C++17, and Fortran 2018

SPECspeed Suites:

- Integer benchmarks now parallelized (9 of 13, up from 1 of 10)
- Memory size increased to 64GB, up from 16GB
- Expanded parallel models

Reporting and Rules Changes:

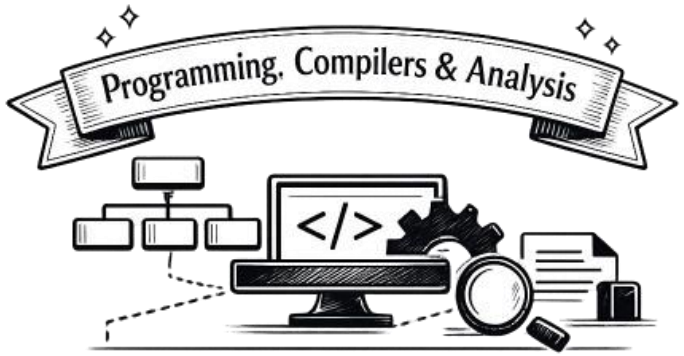
- Improved run rules for relevance of compiler optimizations
- Improved support for bare-metal cloud platforms
- Added report field to indicate use of Open-Source compilers



Technical Analysis Related Changes vs. SPEC CPU 2017

- Many more workloads per application benchmark
  - SPECrate **Integer:** ~2x avg workloads per benchmark (20 to 57 total workloads)
  - SPECrate **Floating Point:** ~4x avg workloads per benchmark (13 to 47 total workloads)
  - Significant increase in workload variety
  - Reduces the overall impact of a single optimization
  - Benchmarks with differing workloads can obfuscate specific microarchitectural behaviors, so studying individual workloads within a benchmark is sometimes important
- Round Robin Rate (RRR)
  - A new Rate scheduling method for experimentation and research
  - Allows for different applications to run at the same time with adjustable staggering
- Shift in CPU core front-end bottleneck
  - Increased code footprint sizes have shifted many front-end bound workloads from branch prediction bound, to instruction cache or instruction translation lookaside buffer (iTLB) bound

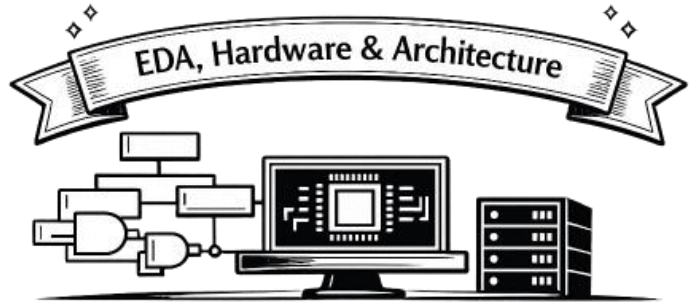
What Are the Benchmarks?



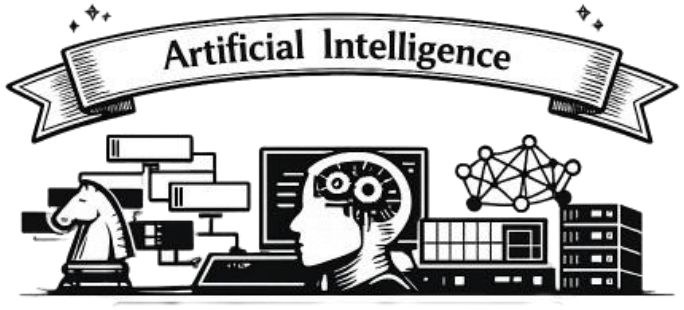
GCC · LLVM · Python · CPPCheck



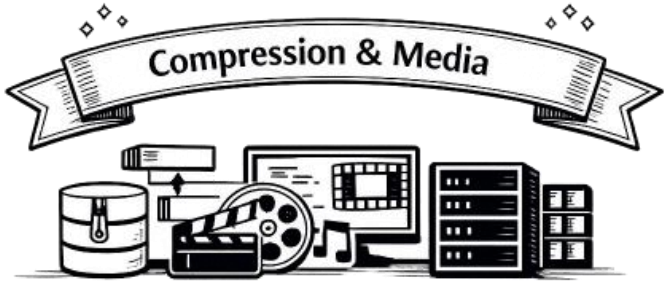
NS3 · Sealcrypto · OMNET++



ABC · VPR · GEM5



Stockfish · NTest · Marian



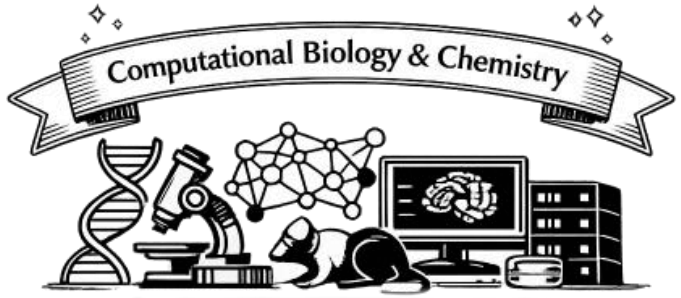
XZ · Zstd · FLAC · astcenc OpenColorIO

Security is a new category included in SPEC CPU 2026 with 750.sealcrypto_r

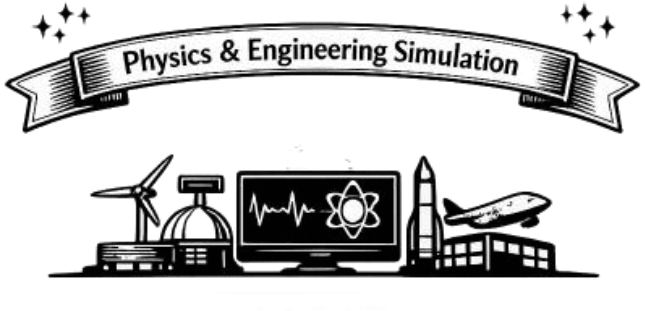
What Are the Benchmarks?



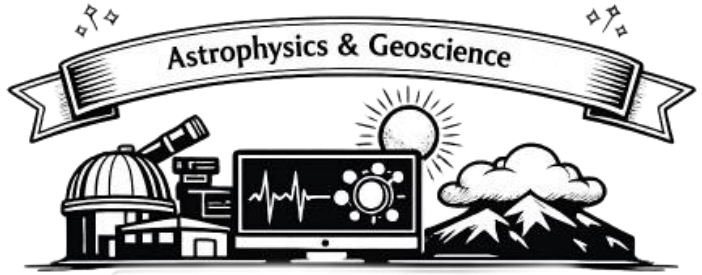
SQLite · Graph500 · MiniZinc



DIAMOND · NAB · NAMD · NEST



Tealeaf · Cloverleaf · FlightDM (JSBSim)
Fotonik3d · FemFlow · LBM · Neutron · Gmsh



PALM · ROMS · POT3D · SPH_EXA CACTUS

Analytics is a new category included in SPEC CPU 2026 with 854.graph500_s

Call to Action

Obtaining SPEC CPU or submitting results:

https://www.spec.org/cpu2026/

Discounts for non-profit organizations, and **free** for qualified academic institutions!

Join the development team! https://www.spec.org/spec/membership/

SPEC is interested in additional partnerships with the OCP community

As OCP and SPEC are both non-profits to improve our industry, collaborations opportunities include performance and power measurement and sustainability

Thank You!

David Reiner

President@SPEC.org

https://www.spec.org/cpu2026/

